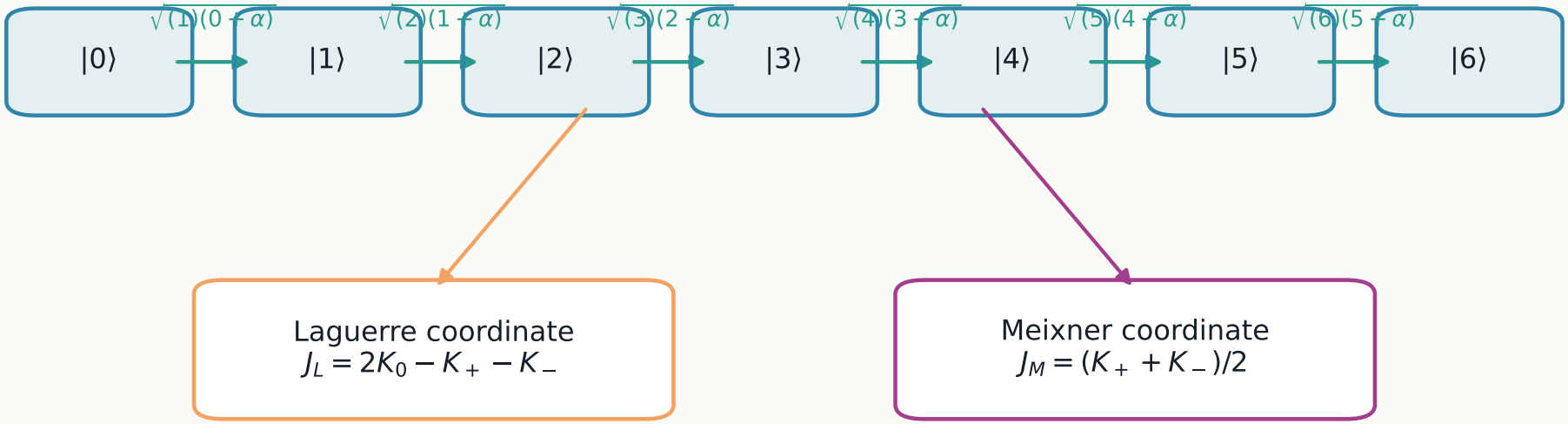
One $\mathfrak{su}(1,1)$ ladder, two orthogonal coordinate systems



The Casimir is $\frac{\alpha}{2}(\frac{\alpha}{2}-1)I$ in both atlases.